

# Experimental demonstration of linear precompensation of a nonlinear transfer function due to second-harmonic generation

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We report on what we believe is the first experimental demonstration of the linear precompensation of a nonlinear transfer function due to frequency conversion. As a proof of principle, we show the effective precompensation with an interferometric filter of FM-to-AM conversion due to second-harmonic generation in a potassium titanyl phosphate crystal. © 2010 Optical Society of America

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The Laser MegaJoule (LMJ) in France and the National Ignition Facility (NIF) in the United States are high-power laser facilities designed to control the implosion of a deuterium–tritium target and to initiate nuclear fusion [1]. Such energetic lasers have to be phase modulated to broaden the optical spectrum in order to suppress stimulated Brillouin scattering and to smooth the speckle pattern illuminating the target [2,3]. In theory, this pure phase modulation does not affect the temporal profile of the laser pulse. Nevertheless, the spectral response of the different optical components is not ideal and thus the FM is partly converted into AM during propagation [4–6]. This conversion adversely affects the laser performance; it can damage the optics or amplify instabilities of the plasma and thus prevent fusion ignition. For the LMJ, intensity modulations due to linear spectral filters can be entirely and easily compensated with inverse transfer functions. On the target, AMs of the input pulse are principally due to nonlinear phenomena in the frequency conversion system (FCS) [4]. So, the suppression of this FM-to-AM conversion due to nonlinear processes is of major interest as evidenced by the recent numerical studies on this topic [7,8]. In a previous paper, we proposed using linear transfer functions to compensate a large part of AMs induced by the FCS [9,10]. In this Letter, we present what we believe to be the first experimental demonstration of this approach. We chose to study the second-harmonic generation of a phase-modulated laser pulse in a potassium titanyl phosphate (KTP) crystal in order to exalt the FM-to-AM conversion process and its compensation.

The laser system is composed of a fiber oscillator that delivers 3 ns flat-in-time laser pulses at 1053 nm, an electro-optic phase modulator, and a regenerative amplifier. We apply a sinusoidal phase modulation, called anti-Brillouin modulation, given by  $\varphi(t) = m \sin(2\pi f_m t)$  with  $m = 7$  and  $f_m = 2$  GHz to create a 28 GHz (0.1 nm) bandwidth spectrum composed of uniformly separated  $f_m$  spaced Dirac peaks, as shown in Fig. 1(a). The phase-modulated pulse is then injected in a regenerative amplifier. At the output of the system, we obtain a 2 mm collimated Gaussian beam with a maximum intensity of around 40 MW/cm<sup>2</sup>. It should be noted that FM-to-AM

conversion in the laser system is negligible. This beam then excites a Type II KTP crystal with a thickness  $L = 2.5$  cm at normal incidence, its intensity  $I$  being adjusted by a variable neutral density filter. After splitting apart the residual IR pulse by a dichroic mirror, we measure the temporal shape of the second-harmonic pulse (526.5 nm) with a photodiode of 150  $\mu\text{m}$  in diameter (only the center of the beam is measured and thus we can neglect the spatial intensity variations) and a 15 GHz bandwidth oscilloscope. The conversion efficiency is around 50%, and if we had perfect phase matching over the whole spectrum width and no spectral filter, the bandwidth spectrum of the second-harmonic pulse would have been 56 GHz, as shown in Fig. 1(b).

In our experiment, the only origin of FM-to-AM conversion is the nonlinear distortion of the optical spectrum due to phase mismatching in the crystal. Indeed, the spectral acceptance  $\gamma$  of the KTP is limited due to an important group velocity difference, which generates amplitude modulations. We have shown in [4] that, to describe the distortion of the optical spectrum, we can consider that the function

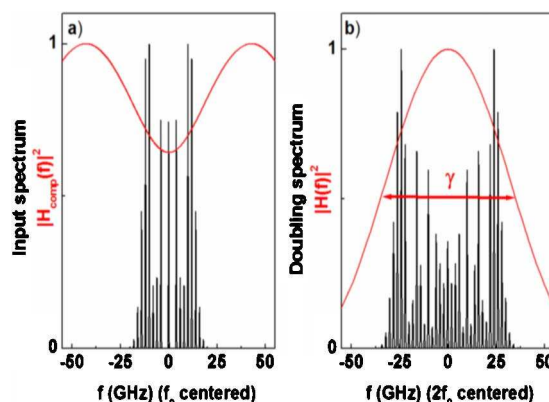


Fig. 1. (Color online) Precompensation of FM-to-AM conversion in the spectral domain. (a) Linear transfer function, which filters the phase-modulated optical spectrum. (b) Transfer function due to the KTP at 10 MW/cm<sup>2</sup>, which filters the spectrum obtained assuming perfect phase-matching condition.

$$H(f) = \text{sinc}\left[\frac{Af}{\gamma}\right] \quad (1)$$

is the filtering function, as though the phenomenon was linear. As a matter of fact, a monochromatic wave at a frequency shifted by  $f$  from the central frequency  $f_0$ , at which there is supposed to be a perfect phase matching, has a reduced amplitude. The parameter  $A$  is a constant fixed to 2.783 in order to define  $\gamma$  as the FWHM of  $|H(f)|^2$  [Fig. 1(b)]. This model is linear and valid for a monochromatic wave assuming no depletion of the fundamental wave during its propagation in the crystal. Nevertheless, the FM-to-AM conversion effect depends on the intensity of the input pulse because  $\gamma$  decreases when  $I$  increases. So, if there are intensity variations,  $H(f)$  varies along the pulse. This simple approach, introduced in [9,10], is theoretically not suitable, but we will show in the following its relevance. We determined the evolution of  $\gamma$  with  $I$  using the CEA's Miró propagation code [11]. As can be seen in Fig. 2(a), when  $I$  varies from 0 to 40 MW/cm<sup>2</sup>,  $\gamma$  decreases from 82 to 34 GHz. To evaluate the resulting modulations, the following distortion criterion  $\alpha$  is commonly introduced:

$$\alpha = 2 \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \quad (2)$$

where  $I_{\max}$  and  $I_{\min}$  represent the maximum and minimum of pulse intensities, respectively. This parameter ranges from 0% to 200% and is ideally equal to 0 (no AM). We measured amplitude modulations of the second-harmonic pulse due to limited spectral acceptance:  $\alpha$  increases from 33% to 130% when  $I$  varies from 0 to 40 MW/cm<sup>2</sup>. These results are very well reproduced by our linear approach, especially at low intensity [Fig. 4(a)]. This approach is not intuitive and is not supposed to give as good results. At high intensity, the results are slightly worse because our model does not take into account the instantaneous variations of  $\gamma$  due to the depletion of the fundamental wave. All these results are also in very good agreement with those obtained by solving the nonlinear coupled-wave equations for the fundamental and second-harmonic pulses with Miró [Fig. 4(a)] [12]. Concerning the spectrum of amplitude

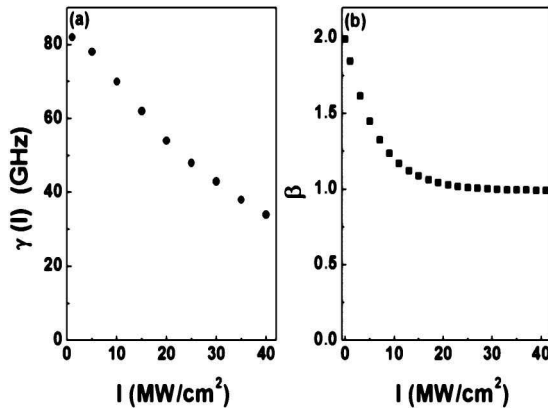


Fig. 2. (a) Simulation of the evolution of the spectral acceptance  $\gamma$  of the KTP crystal with  $I$ . (b) Evolution of  $\beta$  with  $I$ , expressing saturation of second-harmonic generation.

modulations, when the phase-matching condition is satisfied for the central wavelength,  $H(f)$  is centered on the spectrum [as presented in Fig. 1(b)] and thus only harmonics of the frequency modulation  $f_m$ , mostly  $2f_m$  (in our experiment 4 GHz), can be observed (there is no AM at  $f_m$ ) [4].

We will now focus our attention on the reduction of this FM-to-AM conversion effect. Although this process is highly nonlinear, we propose a linear spectral interferometric filter to precompensate a large part of amplitude modulations. We insert this filter in the fibered part of the laser system before amplification for two reasons: it is easier to act on the optical signal when it is less energetic and the regenerative amplifier recovers the energy loss due to the filter. This filter is composed of a birefringent monomode polarization maintaining fiber (PMF), a half-wave plate to set the state of polarization at an angle  $\delta$  of the low axis of the PMF, a fiber polarizer, and a second half-wave plate to set the polarization back to the initial state, as can be seen in Fig. 3 [13]. The PMF length  $L_f$  determines the delay between the two waves propagating in the fiber and the polarizer makes them interfere. The expression of the transfer function of this filter, applying on the fundamental pulse, is the following [9,10]:

$$H_{\text{comp}}(f) = (\cos^2 \delta)(1 + (\tan^2 \delta) \exp[i(2\pi\Delta\tau f + \psi)]), \quad (3)$$

where  $\Delta\tau = L_f \Delta n / c$  is the time delay between the two waves ( $c$  is the velocity of light in vacuum and the birefringence  $\Delta n = 0.5 \times 10^{-3}$ ) and  $\psi$  is the phase recombination determined by constraints applied on the fiber. This is almost a sinusoidal function with a period, an amplitude, and a phase that are, respectively, controlled by  $L_f$ ,  $\delta$ , and  $\psi$ . In our experiment, as the phase-matching condition is satisfied for the central wavelength,  $H_{\text{comp}}(f)$  must be centered on the optical spectrum and, thus, the parameter  $\psi$  must be equal to  $\pi$ . Experimentally, we add a Peltier device controlling temperature of the system with an accuracy of 0.01 °C to precisely adjust  $\psi$ . We chose to design our filter to demonstrate precompensation of FM-to-AM conversion for  $I$  of around 10 MW/cm<sup>2</sup>. For our 7-m-long PMF,  $\Delta\tau$  is about 11.6 ps (no chromatic dispersion) and thus the period is fixed to 86 GHz.  $\delta$  is the optimization parameter and we set its value to 25°. The resulting linear transfer function, which filters the input spectrum, is presented in Fig. 1(a). We demonstrate good agreement with simulations the precompensation of a large part of FM-to-AM conversion: the initial distortion criterion is around

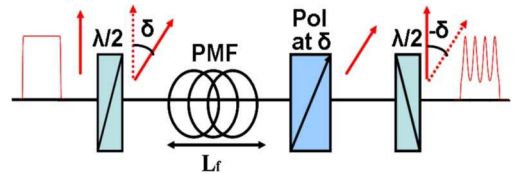


Fig. 3. (Color online) Schematic of the interferometric filter. The evolution of the polarization is represented by an arrow. (Pol, polarizer;  $\lambda/2$ , half-wave plate; PMF, polarization maintaining fiber). A Peltier device is used to control temperature of the system.

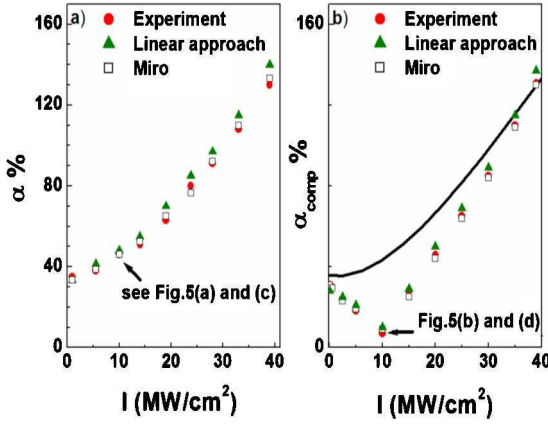


Fig. 4. (Color online) (a) Evolution of the distortion criterion  $\alpha$  of the second-harmonic pulse with  $I$ : comparison among experiment, linear approach using Eq. (1), and nonlinear simulations with Miró. (b) Evolution of  $\alpha_{\text{comp}}$  versus  $I$  with a precompensation system optimized at 10 MW/cm<sup>2</sup>. The solid curve represents the initial evolution of  $\alpha$  shown in (a).

52% and, with our linear filter,  $\alpha_{\text{comp}} < 6\%$ , as can be observed in Fig. 5.

These results can be also interpreted in the temporal domain. The interferometric filter introduces AMs at 4 GHz, and they are amplified by the nonlinearity of frequency conversion by a factor  $\beta$  defined by  $I(2f) \propto I^\beta$  (with  $I(2f)$  as the second-harmonic pulse intensity), this parameter being intensity-dependent because of frequency conversion saturation [9,10]. The evolution of this parameter with  $I$ , numerically determined with Miró, is presented in Fig. 2(b) ( $\beta \leq 2$ ). The resulting modulations are out of phase, with the same amplitude and same frequency as those induced by  $H(f)$  and thus they cancel each other. In that case, there are intensity variations of the input pulse and thus  $H(f)$  varies along the pulse. Therefore, we obtain a precompensation of a large part of AMs but not  $\alpha_{\text{comp}} = 0$ : a linear filter cannot compensate perfectly the instantaneous variations of  $\gamma$  and  $\beta$  with  $I$ . This is the main limitation of our approach: the precompensation is maximized for a unique input intensity. As can be seen in Fig. 4(b), we obtain an excellent precompensation at 10 MW/cm<sup>2</sup> but at low (1 MW/cm<sup>2</sup>) and high intensity (40 MW/cm<sup>2</sup>), AMs are the same with and without compensation. Nevertheless, reduction of FM-to-AM conversion at another intensity can be easily achieved with our experimental setup by only changing the value of  $\delta$  [9,10]. For example, we observed a large compensation at 1 MW/cm<sup>2</sup> ( $\alpha_{\text{comp}} < 5\%$ ) with  $\delta = 21^\circ$ . At high intensity,  $\alpha$  becomes important and thus there are important instantaneous variations of  $\gamma$  and  $\beta$  with  $I$ , but the AMs can still be divided by a factor of 2, in agreement with [9,10].

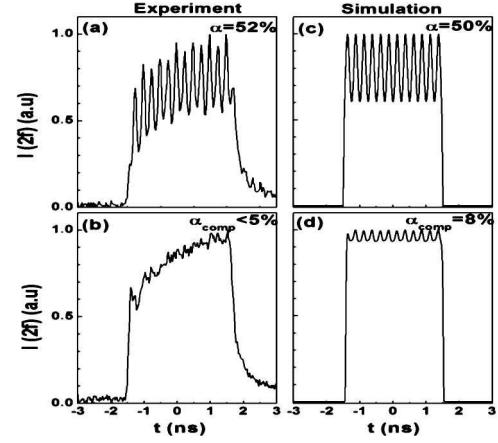


Fig. 5. (a) Experimental measurement of the second-harmonic pulse without compensation for an input intensity of around 10 MW/cm<sup>2</sup>:  $\alpha = 52\%$ . (b) Measurement of intensity modulations with optimized precompensation system:  $\alpha_{\text{comp}} < 6\%$ . Panels (c) and (d), respectively, correspond to the simulations with Miró of panels (a) and (b). The frequency of AM is 4 GHz.

In conclusion, we have demonstrated a simple and innovative approach to reduce FM-to-AM conversion in the LMJ.

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